

Standardization and Generalization

INFO/STSCI/ILRST 3900: Causal Inference

9 Sep 2025

Learning goals for today

At the end of class, you will be able to:

1. Reason about treatment effect heterogeneity
2. Estimate the average causal effect using data from a conditionally randomized experiment

Logistics

- ▶ Ch 2.3 and 4.1-4.3 in Hernan and Robins
- ▶ Quiz 1: Sep 17 in class
 - ▶ Practice questions posted later this week
 - ▶ Review in discussion next week

Exchangeability

$$\text{Average Treatment Effect} = \underbrace{E(Y^{a=1})}_{\text{Avg when all treated}} - \underbrace{E(Y^{a=10})}_{\text{Avg when no one is treated}}$$

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Exchangeability: potential outcomes are independent of treatment status

► **Treated and untreated groups can be swapped:**

Average potential outcome under treatment is the same for actually treated group and actually untreated group:

$$E(Y^{a=1} \mid A_i = 1) = E(Y^{a=1} \mid A_i = 0) = E(Y^{a=1})$$

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► **Treated potential outcomes are observed:**

For the group that is actually treated, I observe potential outcome under treatment

$$E(Y \mid A_i = 1) = E(Y^{a=1} \mid A_i = 1) = E(Y^{a=1})$$

Conditional randomization

Exchangeability may not hold in every randomized experiment

- ▶ In covid vaccine trial, suppose the researchers first split the group into two groups
 - ▶ Older: receive vaccine with probability $2/3$
 - ▶ Younger: receive vaccine with probability $1/4$
- ▶ Does exchangeability still hold?

Conditional randomization

Exchangeability may not hold in every randomized experiment

- ▶ Vaccinated group now contains a larger proportion of older individuals
- ▶ Older individuals more likely to get COVID regardless of treatment

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- ▶ Exchangeability does not hold in entire population

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- ▶ Older individuals more likely to get COVID regardless of treatment
- ▶ Vaccinated individuals have higher proportion of potential outcomes (both $Y^{a=1}$, $Y^{a=0}$) than unvaccinated individuals
- ▶ Exchangeability does not hold in entire population
- ▶ Exchangeability holds within each sub-population
- ▶ Two separate experiments: one for older individuals, one for younger individuals
- ▶ Both experiments are exchangeable

Conditional randomization

- ▶ **Marginal exchangeability:** $Y^a \perp\!\!\!\perp A$ for all a
- ▶ **Conditional exchangeability:** $Y^a \perp\!\!\!\perp A \mid L$ for all a
The potential outcomes are independent of treatment **conditional on L**
- ▶ Roughly speaking: If I only consider a subsets of individuals (categorized by some covariate L), exchangeability holds within the subset

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- ▶ Roughly speaking: If I only consider a subsets of individuals (categorized by some covariate L), exchangeability holds within the subset
- ▶ If you tell me $A_i = 1$, I can predict $Y_i^{a=1}, Y_i^{a=0}$ better than just guessing
- ▶ Suppose you first tell me someone is old, I can predict $(Y_i^{a=1}, Y_i^{a=0})$ better than I could without info.
- ▶ Next you tell me that also $A_i = 1$, I can't improve my prediction of $(Y_i^{a=1}, Y_i^{a=0})$

Conditional Exchangeability

- ▶ Within a group, treated and untreated have some average potential outcomes

$$\begin{aligned} E(Y^{a=1} \mid A_i = 1, L = \text{old}) &= E(Y^{a=1} \mid A_i = 0, L = \text{old}) \\ &= E(Y^{a=1} \mid L = \text{old}) \end{aligned}$$

Stratification

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$$\begin{aligned}\underbrace{E(Y \mid A = 1, L = old)}_{\text{Observe for treated and old}} &= \underbrace{E(Y^{a=1} \mid A = 1, L = old)}_{\text{Within treated and old}} && \text{(Consistency)} \\ &= \underbrace{E(Y^{a=1} \mid A = 0, L = old)}_{\text{Within not treated and old}} && \text{(Conditional Exchangeability)} \\ &= \underbrace{E(Y^{a=1} \mid L = old)}_{\text{everyone}} && \text{(Conditional Exchangeability)}\end{aligned}$$

Conditional randomization

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- ▶ **Stratification:** We can directly estimate causal effect within each sub-population (or stratum)
- ▶ Estimate causal effect for *old*

$$\begin{aligned}\text{ATE for old} &= E(Y^{a=1} \mid L = \textit{old}) - E(Y^{a=0} \mid L = \textit{old}) \\ &= E(Y \mid A = 1, L = \textit{old}) - E(Y \mid A = 0, L = \textit{old})\end{aligned}$$

- ▶ Estimate causal effect for *young*

$$\begin{aligned}\text{ATE for young} &= E(Y^{a=1} \mid L = \textit{young}) - E(Y^{a=0} \mid L = \textit{young}) \\ &= E(Y \mid A = 1, L = \textit{young}) - E(Y \mid A = 0, L = \textit{young})\end{aligned}$$

- ▶ If the treatment effect varies across sub-population, we say there is **treatment effect heterogeneity**

Side note

Average Treatment Effect $E(Y^{a=1}) - E(Y^{a=0})$.1in

Equivalent calculations:

- ▶ Difference of averages
 1. Take average $Y^{a=1}$
 2. Take average $Y^{a=0}$
 3. Take difference between two averages

$$ATE = E(Y^{a=1}) - E(Y^{a=0})$$

Side note

Average Treatment Effect $E(Y^{a=1}) - E(Y^{a=0})$.in

Equivalent calculations:

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1. Take average $Y^{a=1}$
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► Average of differences

1. Get causal effect $Y_i^{a=1} - Y_i^{a=0}$ for every individual
2. Take average of individual causal effects

$$ATE = E(Y^{a=1} - Y^{a=0})$$

Measuring a causal effect

- ▶ Some could have positive causal effect, some could have negative causal effect
- ▶ **Sharp null:** $Y_i^{a=1} - Y_i^{a=0} = 0$ for everyone
i.e., causal effect for every individual is 0
- ▶ Implies $ATE = 0$, but $ATE = 0$ does not imply causal effect for everyone is 0

Measuring a causal effect

- ▶ Some could have positive causal effect, some could have negative causal effect
- ▶ **Sharp null:** $Y_i^{a=1} - Y_i^{a=0} = 0$ for everyone
i.e., causal effect for every individual is 0
- ▶ Implies $ATE = 0$, but $ATE = 0$ does not imply causal effect for everyone is 0
- ▶ Suppose new policy results in .99 of individuals losing 1 dollar; .01 gains 100 dollars
- ▶ 99% have small negative causal effect, 1% has large positive causal effect
- ▶ $ATE = .01 > 0$, but maybe still not a good idea
- ▶ Conditional ATE (ATE within each sub-population) helps give more information

Standardization

- ▶ Under conditional exchangeability, we can directly estimate the conditional average causal effect for each sub-population
- ▶ Conditional ATE

$$E(Y^{a=1} \mid L = \ell) - E(Y^{a=0} \mid L = \ell)$$

- ▶ When different treatments can be applied to different individuals
- ▶ Ex: medicine, job training program

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- ▶ When different treatments can be applied to different individuals
- ▶ Ex: medicine, job training program
- ▶ ATE

$$E(Y^{a=1}) - E(Y^{a=0})$$

- ▶ When everyone gets the same treatment
- ▶ Ex: Tax policy, schoolwide policy
- ▶ **Standardization** allows us to estimate the ACE by combining estimates from each sub-population

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- ▶ Use law of total expectation
- ▶ Take average across groups, but weight each group according to it's size

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Standardization

- **Standardization** allows us to estimate the ACE by combining estimates from each sub-population

$$E(Y^{a=1}) = \sum_{\ell} \underbrace{E(Y^{a=1} \mid L = \ell)}_{\text{Avg within group}} \times \underbrace{Pr(L = \ell)}_{\text{Size of group}}$$

- For $L = 0, 1$

$$\begin{aligned} E(Y^{a=1}) &= Pr(L = 1)E(Y^{a=1} \mid L = 1) + Pr(L = 0)E(Y^{a=1} \mid L = 0) \\ &= Pr(L = 1)E(Y \mid L = 1, A = 1) \\ &\quad + Pr(L = 0)E(Y \mid L = 0, A = 1) \end{aligned}$$

Exercise

		L	A	Y
1	Rheia	0	0	0
2	Kronos	0	0	1
3	Demeter	0	0	0
4	Hades	0	0	0
5	Hestia	0	1	0
6	Poseidon	0	1	0
7	Hera	0	1	0
8	Zeus	0	1	1

		L	A	Y
9	Artemis	1	0	1
10	Apollo	1	0	1
11	Leto	1	0	0
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13	Athena	1	1	1
14	Hephaestus	1	1	1
15	Aphrodite	1	1	1
16	Polyphemos	1	1	1
17	Persephone	1	1	1
18	Hermes	1	1	0
19	Hebe	1	1	0
20	Dionysus	1	1	0

$$\begin{aligned} E(Y^a) &= Pr(L = 1)E(Y \mid L = 1, A = a) \\ &\quad + Pr(L = 0)E(Y \mid L = 0, A = a) \end{aligned}$$

Excercise

		L	A	Y
1	Rheia	0	0	0
2	Kronos	0	0	1
3	Demeter	0	0	0
4	Hades	0	0	0
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$$E(Y^{a=1}) = Pr(L = 1)E(Y \mid L = 1, A = 1)$$

$$+ Pr(L = 0)E(Y \mid L = 0, A = 1)$$

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$$\begin{aligned} E(Y^{a=1}) &= \underbrace{Pr(L=1)}_{12/20} \underbrace{E(Y \mid L=1, A=1)}_{6/9} \\ &\quad + Pr(L=0)E(Y \mid L=0, A=1) \end{aligned}$$

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1	Rheia	0	0	0
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Generalization (transportability)

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- ▶ Randomized control trials provide convincing evidence for a causal effect
- ▶ Unless the sharp null is true, no such thing as “The” average causal effect
- ▶ If treatment effect is heterogeneous, ACE is defined with respect to a population of interest
- ▶ Ex: most psychology experiments consist primarily of undergraduate students. Results may be accurate for undergraduate students, but do not say much about broader population

Generalization (transportability)

- ▶ Drawing representative sample for RCTs yields more reliable conclusions
- ▶ Transporting effect to different population
 - ▶ Assume conditional ACE is same across populations
 - ▶ Standardization should use population weights in target population

Generalization (transportability)

- ▶ Consider experiment conducted on students in a psychology course at Cornell
- ▶ Let $L = 0$ denote undergraduate and $L = 1$ denote graduate student
- ▶ Suppose in the psychology course $P(L = 0) = .95$ and $P(L = 1) = .05$
- ▶ In an anthropology course, $P(L = 0) = .6$ and $P(L = 1) = .4$
- ▶ Suppose

$$E(Y^1 - Y^0 \mid L = 0) = .2$$

$$E(Y^1 - Y^0 \mid L = 1) = -.4$$

- ▶ ACE for the experiment would be

$$.2 \times .95 + (-.4) \times .05 = 0.17$$

- ▶ ACE in the anthropology course would be

$$.2 \times .6 + (-.4) \times .4 = -0.04$$

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